Problems on Probability

1. Let X be a discrete random variable with the following PMF

$$P_X(x) = \begin{cases} 0.1 \text{ for } x = 0.2 \\ 0.2 \text{ for } x = 0.4 \\ 0.2 \text{ for } x = 0.5 \\ 0.3 \text{ for } x = 0.8 \\ 0.2 \text{ for } x = 1 \\ 0 \text{ otherwise} \end{cases}$$

   1. Find the range R_X of the random variable X.
   2. Find P($X \leq 0.5$)
   3. Find P($0.25<X<0.75$)
   4. Find P($X = 0.2|X<0.6$)

2. Two equal and fair dice are rolled and we observed two numbers X and Y.
   1. Find R_X, R_Y and the PMFs of X and Y.
   2. Find $P(X = 2, Y = 6)$.
   3. Find $P(X>3|Y = 2)$.
   4. If $Z = X + Y$. Find the range and PMF of Z.
   5. Find $P(X = 4|Z = 8)$.

3. In an exam, there were 20 multiple-choice questions. Each question had 44 possible options. A student knew the answer to 10 questions, but the other 10 questions were unknown to him and he chose answers randomly. If the score of the student X is equal to the total number of correct answers, then find out the PMF of X. What is P($X>15$)?

4. The number of students arriving at a college between a time interval is a Poisson random variable. On an average 10 students arrive per hour. Let X be the number of students arriving from 10 am to 11:30 am. What is P($10<X\leq15$)?

5. If we have two independent random variables X and Y such that X~Poisson(α) and Y~Poisson(β). Define a new random variable as $Z = X + Y$. Find out the PMF of Z.

12. There is a box containing 40 white shirts and 60 black shirts. If we choose 10 shirts (without replacement) at random, then find the joint PMF of X and Y where X is the number of white shirts and Y is the number of black shirts.

13. If X and Y are two jointly continuous random variables with joint PDF

$$f_{XY}(x, y) = \begin{cases} 6xy \text{ where } 0 \le x \le 1,\ 0 \le y \le \sqrt{x} \\ 0 \text{ otherwise} \end{cases}$$

    1. Find $f_X(x)$ and $f_Y(y)$.
    2. Are X and Y independent to each other?
    3. Find the conditional PDF of X given $Y = y$, $f_{X|Y}(x|y)$.
    4. Find $E[X|Y = y]$, for $0 \le y \le 1$.
    5. Find $Var(X|Y = y)$, for $0 \le y \le 1$.

14. There are 100 men on a ship. If X_i is the weight of the ith man on the ship and Xi's are independent and identically distributed and also $EX_i = \mu = 170$ and $\sigma_{Xi} = \sigma = 30$. Find the probability that the total weight of the men on the ship exceeds 18,000.

15. Let $X_1, X_2, \ldots\ldots, X_{25}$ are the independent and identically distributed.

    And have the following PMF $P_X(k) = \begin{cases} 0.6\ k = 1 \\ 0.4\ k = -1 \\ 0 \text{ otherwise} \end{cases}$

    If $Y = X_1 + X_2 + \ldots + X_n$, estimate $P(4 \le Y \le 6)$ using central limit theorem.